

Brandon Oltarsh

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Education

Purdue University, Class of 2028

Major: Aerospace Engineering

GPA: 4.0

Portfolio

<https://brandonprog.github.io/brandonsportfolio/>

Experience

Purdue Space Program: Active Controls (August 2024-Present)

Cooling analysis/Engine optimization

- Using Matlab to analyze regenerative cooling of a 500-pound force liquid rocket engine
- Developing a method to analyze the effects of film cooling
- Comparing different materials and alcohol based fuels for efficiency and cooling properties

FRC Robotics – (January 2022-April 2024)

Mechanical Director Team 2383

- Leading workshops and design classes to teach underclassmen important concepts of mechanical design and controllability.
- Maintaining a large lab and machine shop hosting workdays of over 60 people.
- Designing and building large robots using Onshape, Autodesk Inventor, and a CamMaster Cobra for competition and outreach events in a team setting.
- Presenting to industry leaders for outreach awards about the teams efforts to expand STEM to disabled groups.

Robotics Summer Camp-(2022-2024, 4 weeks each summer)

Design Teacher/Senior Counselor

- Teaching children ages 10-14 basic design using Onshape with lectures, challenges, and group events.
- Facilitating pick-up and drop off of hundreds of children every day from multiple summer camps.
- Designing STEM focused games for children to play and robot bases for them to build upon.

Craigs Pantry-(August 2018-May 2024)

Volunteer (Every other Sunday)

- Packaging and delivering food for food-insecure population of Broward County.
- Facilitating large donations to the charity from the local community including hundreds of pounds of food and hundreds of articles of clothing.

Skills

- Siemens Teamcenter
- NX, Autodesk Inventor, Fusion 360, Onshape
- CNC Machining (100+ hours experience on a CAMaster Cobra)
- FDM 3D Printing (experience with many commercial slicers)
- Excel suite, Google suite